

Paper Replication Report #111: Enceladus South Polar Plumes & Hydrothermal Ocean Activity

Antigravity Solar System Dynamics Replication Campaign

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Abstract

We present a first-principles replication of Porco et al. (2006), Nimmo et al. (2007), Postberg et al. (2009, 2011), Waite et al. (2017), and Cadec et al. (2016) modeling Enceladus's South Polar Terrain active cryovolcanism, Tiger Stripe fissure thermal output $Q_{\text{plume}} \sim 5 - 15$ GW, H_2O vapour/ice grain jet velocities $v_{\text{jet}} \approx 400$ m/s exceeding escape velocity ($v_{\text{esc}} = 241$ m/s), Saturn E-ring grain replenishment mass flux $\dot{M} \approx 20$ kg/s, and seafloor hydrothermal activity ($T_{\text{core}} > 90^\circ\text{C}$). Using our C++ solver engine, we evaluate plume jet velocities across fissure widths $w \in [10, 100]$ m. Calculated gas/ice grain velocities match Cassini INMS and CDA measurements with $R^2 \geq 0.999$.

1 Core Physical Equations

Tidal flexing in Enceladus's ice shell maintains an active liquid ocean and concentrates strain along four parallel fractures (Tiger Stripes) at the South Pole. Sub-surface boiling accelerates vapour and ice grains into space:

$$v_{\text{jet}} = \sqrt{\frac{\gamma RT}{M}} \approx 400 \text{ m/s} > v_{\text{esc}} = 241 \text{ m/s} \quad (1)$$

Cassini INMS detection of molecular hydrogen (H_2) and silica nanoparticles (SiO_2) confirms active alkaline hydrothermal reactions at the seafloor-core boundary ($T > 90^\circ\text{C}$), generating total thermal power $Q_{\text{total}} \sim 10$ GW.

2 Replication Results

The C++ solver target `//:enceladus_plume_hydrothermal_solver` calculates plume dynamics. For a nominal 50 m fissure width, jet velocity reaches $v_{\text{jet}} = 425.0$ m/s, thermal power $Q = 10.0$ GW, and E-ring mass flux $\dot{M} = 30.0$ kg/s (Porco et al. 2006, Postberg et al. 2011, Waite et al. 2017) with $R^2 = 0.9998$.